Tadalafil

Tadalafil 5 mg orally once daily

Tadalafil sustains nitric-oxide–cGMP signaling by inhibiting cGMP degradation. For erectile dysfunction and benign prostatic hyperplasia, tadalafil is an established prescription PDE5 inhibitor; improved vascular function and an experimental healthspan benefit are plausible additional applications. [12–14,18] Human biomarker studies, animal mechanisms, and observational cardiovascular findings support these hypotheses, while cerebral perfusion and cognitive outcomes must be assessed separately. [3,9,17,21,24]

💉 💉 💉 Tadalafil 5 mg daily has demonstrated safety in selected men followed for up to two years. Continued tolerability under comparable prescribing conditions is likely. [12,14]

⭐ Plausibly effective for improving healthspan. [3,17,22,24]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

In open-label extensions, 234 and 238 men received tadalafil 5 mg daily for one and two years; 208 and 139 completed follow-up, respectively. No drug-related serious adverse events or clinically meaningful treatment-associated ECG or laboratory abnormalities were reported. The open-label design and substantial attrition in the two-year cohort limit inference about rare or delayed harms. [12]

Dosing

A speculative experimental regimen for healthspan research is

- **Tadalafil: 5 mg orally once daily.**

This is an approved daily dose for BPH and ED/BPH; Selected vascular and biomarker studies support the rationale for its use specifically for healthspan. [12–14,19]

Benefits (Beyond Prostate and ED)

- **Endothelial signaling:** Tadalafil inhibits phosphodiesterase type 5, sustaining cyclic guanosine monophosphate and nitric-oxide signaling and producing vasodilation. Daily tadalafil can improve endothelial function in some human populations. These effects

could plausibly protect vascular-dependent tissues from aspects of age-related decline. [13,14,18,19]

- **Microvascular perfusion:** Tadalafil can recruit skeletal-muscle capillaries in humans, although an acute study in postmenopausal women found no increase in total forearm or adipose-tissue blood flow. [10] A small diabetes trial reported microvascular and selected metabolic improvements despite a neutral primary whole-body insulin-resistance endpoint. [25] In septic rats, renal blood flow improved without improved survival, systemic hypotension, or inflammatory endpoints. [27] Tissue-specific perfusion support and preservation of organ function remain plausible; these findings do not imply uniform improvement across organs or conditions.
- **Cerebral and cognitive support:** In the ETLAS-2 exploratory MRI subset, three months of tadalafil 20 mg daily increased whole-brain cerebral blood flow versus placebo; the adjusted difference was 3.33 mL/100 g/min in 42 evaluable participants. Cerebrovascular reactivity and neurovascular-coupling measures did not improve. [17] A single-dose trial, PASTIS, found no significant primary subcortical-flow benefit. [23] The ETLAS-2 cognitive substudy was neutral overall, with a small slowing in one reaction-time test that may reflect multiple testing. [21] Improved cerebral perfusion at 5 mg daily and longer-term neuroprotection remain plausible hypotheses informed by the differing regimens and animal findings. [2,17,26] A post-hoc analysis of 15 participants with type 2 diabetes receiving tadalafil 20 mg daily for six weeks found lower plasma Aβ40, Aβ42, and GFAP; the Aβ42/40 ratio, pTau217, NfL, and GDF-15 did not change significantly. These small biomarker findings suggest a possible biological effect, not a demonstrated cognitive benefit or brain-amyloid clearance. [28]
- **Vascular inflammation and platelet signaling:** Human studies report favorable changes in selected endothelial and inflammatory biomarkers. [13] In a 19-patient pilot study, the platelet P-selectin reduction occurred only in the subgroup without cardiovascular risk factors. [11] A reduction in atherothrombotic events is possible from the combined mechanistic and observational evidence. [3,22]
- **Cardiovascular outcomes:** In a retrospective cohort of men with erectile dysfunction, tadalafil exposure was associated with lower MACE and all-cause mortality. [3] A causal benefit is possible, but prescription-based exposure, residual confounding, and differences in health or sexual activity constrain inference. Broader PDE5-inhibitor cohort meta-analysis supports a class association rather than a tadalafil-specific randomized outcome. [22]
- **Lipid-related biomarkers:** In a 12-week randomized trial in sexually active men with benign prostatic hyperplasia and erectile dysfunction, tadalafil at 2.5 or 5 mg daily

improved lipid-related biomarkers, including the atherogenic index of plasma, with greater improvement at 5 mg. A resulting reduction in cardiovascular events is possible. [13]

- **Lymphatic signaling:** Tadalafil could plausibly modulate lymphatic smooth-muscle tone through cGMP signaling. Isolated rat thoracic-duct experiments demonstrate cGMP/PKG-dependent relaxation, which may reduce resistance while suppressing active pumping. [15] Reports of lymphatic-malformation regression in children concern sildenafil, providing adjacent class evidence rather than direct tadalafil evidence. [16]
- **Mitochondrial protection and autophagy:** In diabetic mouse hearts, tadalafil improved mitochondrial respiration and reduced reactive-oxygen-species production, alongside NO/SIRT1/PGC-1α, eNOS, Akt, and AMPK signaling changes. [24] In an induced Alzheimer-like mouse model, tadalafil modified PI3K/Akt/GSK-3β, PP2A, mTOR/autophagy, cGMP/PKG, and Wnt-related pathways, with increased BDNF and reduced inflammatory signaling. [26] Corresponding cellular protection in humans is plausible.

How These Map to Healthspan

Vascular dysfunction contributes to ischemia, inflammation, and organ impairment during aging. Sustaining NO–cGMP signaling plausibly targets part of this process. Biomarker improvement and animal organ protection support the rationale, while observed cardiovascular associations and cerebral-flow changes remain distinct from demonstrated extension of human healthspan. [3,17,22,24]

Cautions

- **Nitrates or guanylate-cyclase stimulators:** Tadalafil is contraindicated with nitrates in any form and with guanylate-cyclase stimulators such as riociguat because of the risk of clinically significant hypotension. [14]
- **Cardiovascular or blood-pressure concerns:** Unstable cardiovascular disease and hypotension require assessment. Alpha-blockers, antihypertensives, and substantial alcohol can increase blood-pressure effects; the label does not recommend the alpha-blocker combination for BPH. [14]
- **Strong CYP3A4 inhibitors:** Medications such as ketoconazole and ritonavir increase tadalafil exposure. When used with a strong CYP3A4 inhibitor, once-daily tadalafil should not exceed 2.5 mg daily. [14]

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- **Pregnancy and breastfeeding:** Tadalafil use specifically for healthspan during pregnancy or breastfeeding is unestablished. Use should be limited to clinician-directed treatment for an established medical indication. [14]
- **Renal impairment:** Once-daily tadalafil is not recommended when creatinine clearance is below 30 mL/min or during hemodialysis. Moderate renal impairment may require a lower starting dose, depending on the treatment indication. [14]
- **Hepatic impairment:** Once-daily tadalafil requires caution in mild-to-moderate hepatic impairment and is not recommended in severe hepatic impairment. [14]
- **Known serious hypersensitivity to tadalafil. [14]**

Side effects

- **Headache**, dyspepsia, back pain, myalgia, nasal congestion, and flushing. [14]
- **Uncommon but important reported effects:** Clinically significant hypotension, prolonged erection, sudden visual loss, and sudden hearing loss. An erection lasting more than four hours requires emergency care. Stop tadalafil and seek prompt medical attention for sudden vision or hearing loss. [14]

Sex-Specific Differences in PDE5 Inhibitor Data

Most long-term PDE5-inhibitor outcome cohorts involve men; the cardiovascular meta-analysis was 99.4% male. [22] Tadalafil produces microvascular responses in postmenopausal women, supporting likely physiological activity in women, while broader cardiovascular and healthspan benefits remain possible. [10,25] Tolerability may differ: in the small ETLAS-2 tadalafil arm at 20 mg daily, 6 of 13 women and 4 of 25 men stopped treatment because of side effects. This exploratory imbalance does not establish a sex-specific risk estimate for 5 mg daily. [9]

Biological Mechanisms

Primary vascular / signaling effects

Tadalafil inhibits PDE5-mediated cGMP breakdown and thereby supports NO-dependent smooth-muscle relaxation. [14] Human endothelial responses vary with population, regimen, and measurement. [10,18,19] Cerebral blood-flow improvement after sustained treatment is an observed finding in a selected CSVD subset, whereas microvascular responses elsewhere are tissue-specific. [10,17,25,27] CREB/BDNF signaling, synaptic support, metabolite clearance, and neuroprotection remain mechanistically plausible extrapolations from PDE5-inhibitor and tadalafil models. [1,2,7,26]

Downstream maintenance and protective programs

In diabetic mice, tadalafil enhanced cardiac mitochondrial respiration and NO/SIRT1/PGC-1α signaling while reducing oxidative stress. [24] In a chemically induced Alzheimer-like mouse model, it modulated inflammatory and mTOR/autophagy-related pathways. [26] Human endothelial biomarkers and a restricted platelet subgroup provide complementary evidence. [11,13] Tadalafil's cGMP signaling plausibly modulates lymphatic tone, informed by rat lymphatic physiology and adjacent sildenafil observations. [15,16]

Conclusions

Tadalafil is an established prescription PDE5 inhibitor with a long clinical safety history. Its effects on endothelial function, tissue perfusion, and related protective pathways provide a biologically coherent rationale for a plausible healthspan benefit. Preclinical, human biomarker, and observational findings support this experimental application. [3,12,14,17,22,24]

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